

CAP - Production Capstone - Marketplace Data Platform

Multi-source platform | quality + warehouse + orchestration + Spark/Delta where justified + Fabric design + incidents

Business/domain

OpenSquare marketplace operations

How this book works

This book has exactly 12 phase pages. Each phase has one artifact/evidence contract and one protected rubric page. The small teaching pattern is not the final solution; apply the reasoning to the supplied project workspace yourself.

Project control before Phase 1

- Keep protected Review and sealed failure/incident packs closed until the named phase.
- Create a real project branch/working copy if using Git.
- Record tool/runtime versions only after observing them.
- Do not delete a bad first attempt.
- Use the exact phase evidence ID and path; this keeps Mentor/book/evidence routing one-to-one.

Phase map

CAP-01 - Read capstone requirements and write source contracts

CAP-02 - Design architecture and record major ADRs

CAP-03 - Build multi-source ingestion and raw evidence

CAP-04 - Implement critical quality, quarantine and reconciliation

CAP-05 - Build trusted warehouse/dbt serving layer

CAP-06 - Orchestrate dependencies, retries and bounded backfills

CAP-07 - Use Spark/Delta where scale justifies it

CAP-08 - Design Fabric deployment, access and secret boundaries

CAP-09 - Add CI/CD, monitoring, alerts and runbook

CAP-10 - Open incident pack, diagnose and recover safely

CAP-11 - Reproduce from clean state and freeze evidence bundle

CAP-12 - Final technical defense and Gate 7 readiness

CAP-01 - Read capstone requirements and write source contracts

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete read capstone requirements and write source contracts and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Source contracts are the capstone foundation. API, CSV and database sources need grain, key, timestamps, required fields, duplicate/redelivery and schema-change rules.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Write one contract per source. If you cannot say what one row/object means, do not design the Gold model yet.

BUILD THIS ARTIFACT

source_contracts.md

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

Add/remove/change one field according to supplied compatibility rule; quarantine/fail safely rather than silently corrupt.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-01-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-01/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-02.

CAP-02 - Design architecture and record major ADRs

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete design architecture and record major adrs and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Architecture decisions must be traceable to requirements. Record ADRs for important choices and rejected alternatives.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Separate raw, validation/quarantine, trusted transform, serving, orchestration, monitoring and security boundaries.

BUILD THIS ARTIFACT

architecture.md/diagram + ADRs

VALIDATE

Trace every source to trusted/serving output; identify one failure boundary and replay/recovery path.

CHANGED / FAILURE TEST

PHASE CHECK CAP-02: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-02-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-02/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-03.

CAP-03 - Build multi-source ingestion and raw evidence

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete build multi-source ingestion and raw evidence and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Multi-source ingestion must preserve replayable raw evidence and partial-failure truth. A paginated API is incomplete until all required pages are durable/validated.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Persist each API page/file/snapshot with source identity/hash/run metadata; never silently call partial success complete.

BUILD THIS ARTIFACT

raw/landing evidence + run_metadata.json

VALIDATE

Source rows/events reconcile to raw + quarantine + deliberate duplicate handling; raw evidence remains replayable.

CHANGED / FAILURE TEST

PHASE CHECK CAP-03: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-03-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-03/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-04.

CAP-04 - Implement critical quality, quarantine and reconciliation

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete implement critical quality, quarantine and reconciliation and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Critical quality rules decide publish eligibility. Reconcile source -> accepted/quarantine/dedup and independent business totals.

TINY PATTERN - NOT YOUR FINAL SOLUTION

A required key, duplicate identity, relationship or amount defect must have explicit failure action and evidence.

BUILD THIS ARTIFACT

tests/ + reconciliation report + quarantine evidence

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK CAP-04: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-04-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-04/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-05.

CAP-05 - Build trusted warehouse/dbt serving layer

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete build trusted warehouse/dbt serving layer and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Trusted serving layer starts from grain and lineage. SQL/dbt is a means; correctness comes from keys, tests and reconciliation.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Build daily seller revenue and delivery exceptions only after defining one-row meaning and join cardinality.

BUILD THIS ARTIFACT

models/SQL/dbt + lineage + row-count controls

VALIDATE

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

CHANGED / FAILURE TEST

PHASE CHECK CAP-05: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-05-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-05/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-06.

CAP-06 - Orchestrate dependencies, retries and bounded backfills

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete orchestrate dependencies, retries and bounded backfills and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Orchestration/backfill design must protect later state. Retries are safe only when tasks/side effects are idempotent or controlled.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Represent ingest -> quality -> transform -> publish -> monitor; show a bounded historical repair without rewriting newer truth.

BUILD THIS ARTIFACT

cap_phase_artifact + evidence note

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK CAP-06: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-06-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-06/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-07.

CAP-07 - Use Spark/Delta where scale justifies it

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete use spark/delta where scale justifies it and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Use Spark/Delta only where scale/data-shape justifies it. Extra distributed technology is not a badge.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Write the threshold/shape that makes local/SQL processing insufficient, then show how Spark/Delta responsibility changes.

BUILD THIS ARTIFACT

Spark job/notebook + explain plan + partition/performance evidence

VALIDATE

Compare output to small trusted control result; inspect plan/partitions; justify performance choice without changing correctness.

CHANGED / FAILURE TEST

Supply skewed/unbalanced input and require correctness-preserving diagnosis; optimization is secondary to invariant safety.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-07-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-07/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-08.

CAP-08 - Design Fabric deployment, access and secret boundaries

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete design fabric deployment, access and secret boundaries and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Fabric design must show identity/access/secret boundaries. Do not claim a tenant run if only a design was completed.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Map OneLake/Lakehouse/warehouse/pipeline/monitoring plus least-privilege roles and secret location.

BUILD THIS ARTIFACT

Fabric deployment design + connection/access/secret boundary notes

VALIDATE

Map identities/connections/access/secret boundaries; prove no plaintext secret or broad unsafe access is required.

CHANGED / FAILURE TEST

PHASE CHECK CAP-08: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-08-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-08/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-09.

CAP-09 - Add CI/CD, monitoring, alerts and runbook

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete add ci/cd, monitoring, alerts and runbook and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

CI/CD and observability protect change and operation. Monitoring must cover freshness, quality, failure and cost/capacity signals.

TINY PATTERN - NOT YOUR FINAL SOLUTION

A green unit test is not enough: include publish gate, secret scan/policy, deployment rollback and stale-data response.

BUILD THIS ARTIFACT

README.md + runbook.md + evidence_index.md

VALIDATE

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

CHANGED / FAILURE TEST

PHASE CHECK CAP-09: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-09-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-09/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-10.

CAP-10 - Open incident pack, diagnose and recover safely

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete open incident pack, diagnose and recover safely and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Open the incident pack only now. Work one incident at a time so first-failure evidence and root cause are attributable.

TINY PATTERN - NOT YOUR FINAL SOLUTION

For each incident: reproduce -> preserve failure -> diagnose -> smallest repair -> fresh retry -> regression controls.

BUILD THIS ARTIFACT

first-failure evidence + root_cause.md + repair + fresh retry evidence

VALIDATE

Save first failing evidence before repair; identify root cause; apply smallest safe repair; fresh retry must pass all prior controls.

CHANGED / FAILURE TEST

MANDATORY SUPPLIED FAILURE: capture first failure, diagnose, repair, fresh retry; do not overwrite original evidence.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-10-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-10/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-11.

CAP-11 - Reproduce from clean state and freeze evidence bundle

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete reproduce from clean state and freeze evidence bundle and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Clean-state reproduction proves the project is transferable. Freeze evidence rather than editing the original failed attempt out of history.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Run from documented clean state, replay same input, add one changed/late record, and verify only intended state changes.

BUILD THIS ARTIFACT

state ledger/watermark + rerun/backfill evidence

VALIDATE

Run same input twice and prove trusted logical result is stable; then add a changed/new record and prove only intended state changes.

CHANGED / FAILURE TEST

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-11-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-11/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: CAP-12.

CAP-12 - Final technical defense and Gate 7 readiness

STARTING STATE

Open the project workspace and your last saved evidence. Keep protected Review and any sealed incident/failure pack closed unless this phase explicitly unlocks it.

OUTCOME

Independently complete final technical defense and gate 7 readiness and produce reviewable evidence that proves the result, not merely the action.

TEACHER EXPLANATION

Final defense is a technical conversation, not a memorized speech. Be able to change one requirement and revise architecture on the spot.

TINY PATTERN - NOT YOUR FINAL SOLUTION

Defend business problem, grain, architecture, quality, rerun/backfill, one incident, security, cost/scale and honest limitations.

BUILD THIS ARTIFACT

defense_notes.md + reviewer checklist

VALIDATE

Reviewer can locate every evidence artifact and challenge one design trade-off; learner explains limitation and recovery path without reading a script.

CHANGED / FAILURE TEST

PHASE CHECK CAP-12: one changed-input or broken-assumption test must be run and documented before mastery.

EVIDENCE + REVIEW ROUTE

Evidence ID: M18-CAP-12-EVIDENCE. Save:

03_PROJECT_WORKSPACES/CAP_PRODUCTION_MARKETPLACE_PLATFORM/evidence/CAP-12/attempt.json. CAP rubric -> exact phase criteria; review only after attempt evidence exists.

MASTER / MOVE-ON

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

STOP - EXPLAIN BACK

Without reading this page, explain what changed in the system, what evidence proves it, one way it could be wrong, and what your changed/failure test taught you. Next: M19 DE19-01.